6.1 The Benamou-Brenier formula and its numerical applications

The results of Sections 5.3 and 5.4 allow to rewrite the optimization problem corresponding to the cost $|x-y|^p$ in a smart way, so as to transform it into a convex optimization problem. Indeed

- looking for an optimal transport for the cost $c(x,y)=|x-y|^p$ is equivalent to looking for constant speed geodesic in $\mathbb{W}_p$ because from optimal plans we can reconstruct geodesics and from geodesics (via their velocity field) it is possible to reconstruct the optimal transport;
- constant speed geodesics may be found by minimizing $\int_0^1 |\mu'|(t)^p \mathrm{d}t$;
- in the case of the Wasserstein spaces, we have $|\mu'|(t)^p = \int_\Omega |\mathbf{v}_t|^p \,\mathrm{d}\mu_t$, where $\mathbf{v}$ is a velocity field solving the continuity equation together with μ (this field is not unique, but the metric derivative $|\mu'|(t)$ equals the minimal value of the L^p norm of all possible fields).

As a consequence of these last considerations, for $p>1$, solving the kinetic energy minimization problem

$$\min\left\{\int_0^1\int_\Omega |\mathbf{v}_t|^p \mathrm{d}\rho_t \,\mathrm{d}t \ :\ \partial_t\rho_t+\nabla\cdot(\mathbf{v}_t\rho_t)=0,\ \rho_0=\mu,\ \rho_1=\nu\right\}$$

selects constant speed geodesics connecting μ to ν, and hence allows to find the optimal transport between these two measures[1].

On the other hand, this minimization problem in the variables $(\rho_t,\mathbf{v}_t)$ has non-linear constraints (due to the product $\mathbf{v}_t\rho_t$) and the functional is non-convex (since $(t,x)\mapsto t|x|^p$ is not convex). Yet, we saw with the tools of Section 5.3.1 that it is possible to transform it into a convex problem.

For this, it is sufficient to switch variables, from $(\rho_t,\mathbf{v}_t)$ into (ρ_t,E_t) where $E_t=\mathbf{v}_t\rho_t$ and use the functional $\mathscr{B}_p$ in space-time. Remember $\mathscr{B}_p(\rho,E):=\int f_p(\rho,E)$ with $f_p:\mathbb{R}\times\mathbb{R}^d\to\mathbb{R}\cup\{+\infty\}$ defined in Lemma 5.17. We recall its definition for the reader's convenience:

$$f_p(t,x):=\sup_{(a,b)\in K_q}(at+b\cdot x)=\begin{cases}\frac{1}{p}\frac{|x|^p}{t^{p-1}} & \text{if } t>0,\\ 0 & \text{if } t=0, x=0\\ +\infty & \text{if } t=0, x\neq 0,\ \text{ or } t<0,\end{cases}$$

where $K_q:=\{(a,b)\in\mathbb{R}\times\mathbb{R}^d\ :\ a+\frac{1}{q}|b|^q\le 0\}$. The problem becomes:

Problem 6.1. Solve

$$(\mathrm{B}_p\mathrm{P})\qquad \min\left\{\mathscr{B}_p(\rho,E)\ :\ \partial_t\rho_t+\nabla\cdot E_t=0,\ \rho_0=\mu,\ \rho_1=\nu\right\}.$$

[1] We choose to denote by ρ the interpolating measures, in order to stress the "continuous" flavor of this method (typically, ρ is a density, and μ a generic measure)

Note that we can write $\mathscr{B}_p(\rho,E) = \int_0^1 \mathscr{B}_p(\rho_t,E_t)\mathrm{d}t = \int_0^1 \int_\Omega f_p(\rho_t(x),E_t(x))\,\mathrm{d}x\mathrm{d}t$, where this third expression of the functional implicitly assumes $\rho_t, E_t \ll \mathscr{L}^d$. Indeed, as we saw in Proposition 5.18, the functional $\mathscr{B}_p$ has an integral representation of this form as soon as ρ_t and E_t are absolutely continuous w.r.t. a same positive measure[2].

We also want to stress that the constraints given by $\partial_t\rho_t + \nabla\cdot E_t = 0$, $\rho_0 = \mu$, $\rho_1 = \nu$ are indeed a divergence constraint in space-time (consider the vector $(\rho,E) : [0,1]\times\Omega \to \mathbb{R}^{d+1}$). The space boundary constraints are already of homogeneous Neumann type, while the initial and final value of ρ provide non-homogeneous Neumann data on the boundaries $\{0\}\times\Omega$ and $\{1\}\times\Omega$. Indeed, the whole constraint can be read as $\nabla_{t,x}\cdot(\rho,E) = \delta_0\otimes\mu - \delta_1\otimes\nu$ (notice that this point of view which uses derivatives in (t,x) will appear again in the section). The functional that we minimize is a 1-homogeneous functional, and in this way (B$_p$P) becomes a dynamical version (in space-time) of the Beckmann probelm that we saw in Section 4.2. It is by the way the problem that we obtain if we apply to the cost $|x-y|^p$ the reduction suggested in [197], which transforms it into a one-homogeneous transport cost in space-time.

The constraints are now linear, and the functional convex. Yet, the functional (and the function f_p as well) is convex, but not so much, since, as we said, it is 1-homogeneous. In particular, it is not strictly convex, and not differentiable. This reduces the efficiency of any gradient descent algorithm in order to solve the problem, but some improved methods can be used.

In [36], the authors propose a numerical method, based on this convex change of varibales, on duality, and on what is called "augmented Lagrangian".

Box 6.1. **– Memo** – Saddle points, Uzawa and Augmented Lagrangian

Suppose that we need to minimize a convex function $f:\mathbb{R}^N\to\mathbb{R}$, subject to k linear equality constraints $Ax = b$ (with $b\in\mathbb{R}^k$ and $A\in \mathrm{M}^{k\times N}$). This problem is equivalent to

$$\min_{x\in\mathbb{R}^N} f(x) + \sup_{\lambda\in\mathbb{R}^k} \lambda\cdot(Ax-b).$$

This gives a min-max problem with a Lagrangian function $L(x,\lambda) := f(x)+\lambda\cdot(Ax-b)$. If we believe in duality, finding a minimizer for f under the constraints is the same of finding a maximizer for $g(\lambda) := \inf_{x\in\mathbb{R}^N} f(x)+\lambda\cdot(Ax-b)$. And it is the same as finding a saddle point for L (a point $(\bar x,\bar\lambda)$ where $L(x,\bar\lambda)\ge L(\bar x,\bar\lambda)\ge L(\bar x,\lambda)$ for every x and every λ, i.e. a point which minimizes in x and maximizes in λ).

The maximization in λ is easier, since the problem is unconstrained: it is possible to apply a gradient algorithm (see Box 6.8). We only need to compute $\nabla g(\lambda)$, but this is easily given via $\nabla g(\lambda) = Ax(\lambda)-b$, where $x(\lambda)$ is the (hopefully unique) minimizer of $x\mapsto f(x)+\lambda\cdot(Ax-b)$.

The algorithm that we obtain from this idea, called *Uzawa algorithm*[3], reads as follows: given (x_k,λ_k), set $x_{k+1} := x(\lambda_k)$ and $\lambda_{k+1} = \lambda_k + \tau\nabla g(\lambda_k) = \lambda_k + \tau(Ax_{k+1}-b)$, for a given small τ. The sequence of points x_k converges, under reasonable assumptions, to the minimizer of the original problem.

[2] This is typical of 1-homogeneous functionals (the fact that the result is independent of the reference measure).

An alternative idea, which makes the computation of the point $x(\lambda)$ easier and accelerates the convergence is the following: instead of using the Lagrangian $L(x,\lambda)$, use the following variant: $\widetilde{L}(x,\lambda) := L(x,\lambda) + \frac{\widetilde{\tau}}{2}|Ax-b|^2$, for a given value of $\widetilde{\tau}$. The conditions for being a saddle point of $\widetilde{L}$ and L are, respectively,

$$\text{for } \widetilde{L}: \begin{cases} \nabla f(x) + A^t\lambda + \widetilde{\tau}(Ax-b) = 0, \\ Ax-b=0, \end{cases} \qquad \text{for } L: \begin{cases} \nabla f(x) + A^t\lambda = 0, \\ Ax-b=0. \end{cases}$$

Hence, the saddle points of $\widetilde{L}$ and L are the same, but using $\widetilde{L}$ makes the problem more convex and tractable in x. Then, one uses the same gradient algorithm on $\widetilde{g}(\lambda) := \inf_x f(x) + \lambda\cdot(Ax-b) + \frac{\widetilde{\tau}}{2}|Ax-b|^2$, using a step τ. It is possible to take $\tau = \widetilde{\tau}$ and iterate the following algorithm:

$$\begin{cases} x_{k+1} = \operatorname{argmin} f(x) + \lambda_k\cdot(Ax-b) + \frac{\tau}{2}|Ax-b|^2 \\ \lambda_{k+1} = \lambda_k + \tau(Ax_{k+1}-b) \end{cases}.$$

For more details and convergence result, we refer for instance to [171].

Here are the main steps to conceive the algorithm.

First of all, we will write the constraint in a weak form (actually, in the sense of distributions), thanks to (4.3). This means that we actually want to solve

$$\min_{\rho,E} \quad \mathscr{B}_p(\rho,E) + \sup_\phi \left(-\int_0^1 \int_\Omega ((\partial_t\phi)\,\rho_t + \nabla\phi\cdot E_t) + G(\phi) \right),$$

where we set

$$G(\phi) := \int_\Omega \phi(1,x)\,\mathrm{d}\nu(x) - \int_\Omega \phi(0,x)\,\mathrm{d}\mu(x),$$

and the sup is computed over all functions defined on $[0,1]\times\Omega$ (we do not care here about their regularity, since they will be anyway represented by functions defined on the points on a grid in $[0,1]\times\mathbb{R}^d$).

Remark 6.2. It is interesting to make a small digression to see the connection between the above problem and a Hamilton-Jacobi equation. We will consider the easiest case, $p=2$. In this case we can write the problem as

$$\min_{(E,\rho):\rho\geq 0} \int_0^1 \int_\Omega \frac{|E|^2}{2\rho} + \sup_\phi - \int_0^1 \int_\Omega ((\partial_t\phi)\,\rho + \nabla\phi\cdot E) + G(\phi),$$

where we expressed the functional $\mathscr{B}_2$ with its integral expression, valid in the case of absolutely continuous measures, with $\rho \geq 0$ (where $\rho = 0$, we must have $E=0$ in order to have finite energy).

If we formally interchange inf and sup, we get the following problem

$$\sup_\phi \quad G(\phi) + \inf_{(E,\rho):\rho\geq 0} \int_0^1 \int_\Omega \left(\frac{|E|^2}{2\rho} - (\partial_t\phi)\,\rho - \nabla\phi\cdot E \right).$$

We can first compute the optimal E, for fixed ρ and ϕ, thus getting $E = \rho\nabla\phi$. The problem becomes

$$\sup_{\phi} \quad G(\phi) + \inf_{\rho \geq 0} \int_0^1 \int_\Omega \left(-(\partial_t \phi)\rho - \frac{1}{2}|\nabla\phi|^2\rho \right).$$

The condition for the infimum to be finite (and hence to vanish) is $\partial_t\phi + \frac{1}{2}|\nabla\phi|^2 \leq 0$, and at the optimum we must have equality on $\{\rho > 0\}$. This gives the Hamilton-Jacobi equation

$$\partial_t\phi + \frac{1}{2}|\nabla\phi|^2 = 0 \quad \rho\text{-a.e.}$$

By the way, from the optimal ϕ we can recover the Kantorovich potentials using $\psi(x) := \phi(1,x)$ and $\varphi(x) := -\phi(0,x)$.

The quantity $\mathscr{B}_p$ in this variational problem may be expressed as a sup and hence we get

$$\min_{\rho,E} \sup_{(a,b)\in K_q,\phi} \int_0^1 \int_\Omega (a(t,x)\mathrm{d}\rho + b(t,x)\cdot \mathrm{d}E - \partial_t\phi\, \mathrm{d}\rho - \nabla\phi\cdot \mathrm{d}E) + G(\phi).$$

Denote $m = (\rho, E)$. Here $m : [0,1]\times\Omega \to \mathbb{R}^{d+1}$ is a $(d+1)$-dimensional vector field defined on a $(d+1)$-dimensional space. Again, we do not care here about m being a measure or a true function, since anyway we will work in a discretized setting, and m will be a function defined on every point of a grid in $[0,1]\times\mathbb{R}^d$. Analogously, we denote $\xi = (a,b)$. We also denote by $\nabla_{t,x}\phi$ the space-time gradient of ϕ, i.e. $\nabla_{t,x}\phi = (\partial_t\phi, \nabla\phi)$.

The problem may be re-written as

$$\min_m \sup_{\xi,\phi\,:\,\xi\in K_q} \langle \xi - \nabla_{t,x}\phi, m\rangle + G(\phi).$$

Here comes the idea of using an augmented lagrangian method. Indeed, the above problem recalls a Lagrangian, but in a reversed way. We must think that the dual variable should be m, and the primal one is the pair (ξ,ϕ). The function $f(\xi,\phi)$ includes the term $G(\phi)$ and the constraint $\xi \in K_q$, and there is an equality constraint $\xi = \nabla_{t,x}\phi$. We do not care actually at the original constrained problem giving rise to this Lagrangian, bur we just decide to add in the optimization a term $\frac{\tau}{2}|\xi - \nabla_{t,x}\phi|^2$ (for a small step size τ).

Hence, we look for a solution of

$$\min_m \sup_{\xi,\phi\,:\,\xi\in K_q} \langle \xi - \nabla_{t,x}\phi, m\rangle + G(\phi) - \frac{\tau}{2}|\xi - \nabla_{t,x}\phi|^2.$$

The algorithm that one can consider to find the optimal m shpuld do the following: produce a sequence m_k and find, for each of them, the optimal (ξ_k, ϕ_k). Yet, instead of finding exactly the optimal (ξ_k, ϕ_k) we will optimize in two steps (first the optimal ϕ for fixed ξ, then the optimal ξ for this ϕ).

The algorithm will work in three iterates steps. Suppose we have a triplet (m_k, ξ_k, ϕ_k)

- Given m_k and ξ_k, find the optimal ϕ_{k+1}, by solving

$$\max_{\phi} \quad -\langle \nabla_{t,x}\phi, m_k\rangle + G(\phi) - \frac{\tau}{2}||\xi_k - \nabla_{t,x}\phi||^2,$$

  which amounts to minimizing a quadratic problem in $\nabla_{t,x}\phi$. The solution can be found as the solution of a Laplace equation $\tau\Delta_{t,x}\phi = \nabla\cdot(\tau\xi_k - m_k)$, with a space-time Laplacian, with Neumann boundary conditions. These conditions are homogeneous in space, and non-homogeneous, due to the role of G, on $t = 0$ and $t = 1$. Most Laplace solvers can find this solution in time $O(N \log N)$, where N is the number of points in the discretization.

- Given m_k and ϕ_{k+1}, find the optimal ξ_{k+1}, by solving

$$\max_{\xi\in K_q} \quad \langle \xi, m_k\rangle - \frac{\tau}{2}|\xi - \nabla_{t,x}\phi_{k+1}|^2.$$

  By expanding the square, we see that this problem is equivalent to the projection of $\nabla_{t,x}\phi_{k+1} + \frac{1}{\tau}m_k$, and no gradient appears in the minimization. This means that the minimization may be performed pointwisely, by selecting for each (t,x) the point $\xi = (a,b)$ which is the closest to $\nabla_{t,x}\phi_{k+1}(t,x) + \frac{1}{\tau}m_k(t,x)$ in the convex set K_q. If we have a method for this projection in $\mathbb{R}^{n+1}$, requiring a constant number of operations, then the cost for this pointwise step is $O(N)$.

- Finally we update m by setting $m_{k+1} = m_k - \tau(\xi_{k+1} - \nabla_{t,x}\phi_{k+1})$.

This algorithm globally requires $O(N \log N)$ operations at each iterations (warning: N is given by the discretization in space-time). It can be proven to converge, even if, for convergence issues, the reader should rather refer to the works by K Guittet and Benamou-Brenier-Guittet, [189, 37, 190], as the original paper by Benamou and Brenier does not insist on this aspect. Also see the recent paper [195] which fixes some points in the proof by Guittet.

Compared to other algorithms, both those that we will present in the rest of the chapter and other more recent ones, the Benamou-Brenier method has some important advantages:

1. it is almost the only one for the moment which takes care with no difficulties of vanishing densities, and does not require special assumptions on their supports;

2. It is not specific to the quadratic cost: we presented it for power costs $c(x,y) = |x-y|^p$ but can be adapted to other convex costs $h(x-y)$ and to all costs issued from a Lagrangian action (see Chapter 7 in [293]); it can incorporate costs depending on the position and of time, and is suitable for Riemaniann manifolds;

3. it can be possibly adapted to take into account convex constraints on the density ρ (for instance lower or upper bounds);

4. it can handle different "dynamical" problems, where penalization on the density or on the velocity are added, as it happens in the case of mean-field games (see Section 8.4.4) and has been exploited, for instance, in [39];

5. it can handle multiple populations, with possibly interacting behaviors.

We refer for instance to [97, 248] for numerical treatments of the variants of the points 2. and 3. above, and to [38] for an implementation of the multi-population case. Here we present a simple picture (Figure 6.1), obtained in the easiest case via the Benamou-Brenier method on a torus.

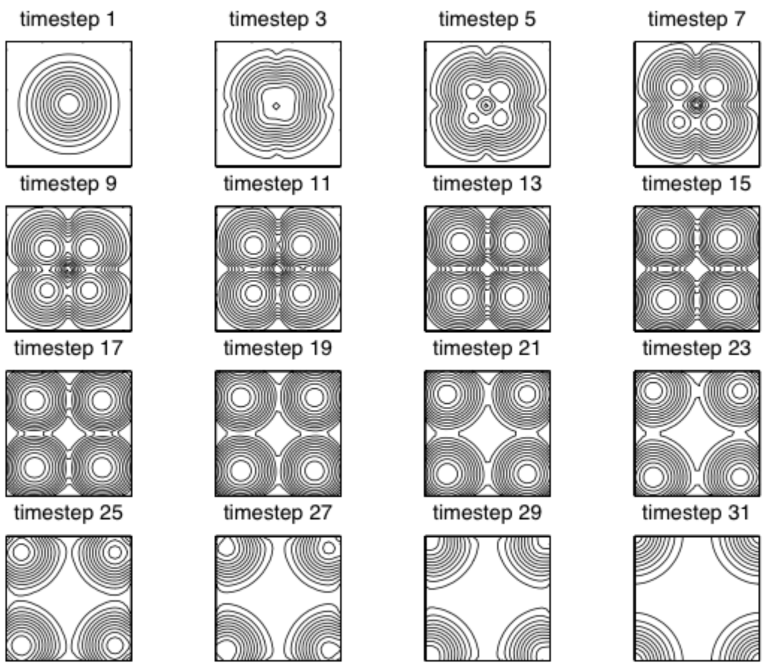

Figure 6.1: Geodesics between a gaussian on a 2D torus, and the same gaussian displaced of a vector $(1/2,1/2)$. Notice that, to avoid the cutlocus (see Section 1.3.2), the initial density breaks into four pieces. Picture taken from [36] with permission.

6.2 Angenent-Haker-Tannenbaum

The algorithm that we see in this section comes from an interesting minimizing flow proposed and studied in [20]. The main idea is to start from an admissible transport map T between a given density f (that we will suppose smooth and strictly positive on a nice compact domain Ω) and a target measure ν (we do not impose any kind of regularity on ν for the moment), and then to rearrange the values of T slowly in time, by composing it with the flow of a vector field $\mathbf{v}$, i.e. considering $\mathrm{T}\circ(Y_t)^{-1}$ instead of